CDAC-Mumbai
Advanced Programming for AI using Python

Date:23-03-2026 **Total Marks:25**

Time duration: 1.30 hours. **Batch: Feb26**

Question: 1 [15 marks]

Design and implement a **Python-based Online Food Ordering System** using **Object Oriented Programming** concepts and **Exception Handling**.

Functional Requirements

1. Create an **abstract base class** `FoodItem` with:
   - attributes: `item_id`, `name`, `price`
   - an abstract method `calculate_total()`

2. Implement **Encapsulation**:
   - Make price and item details private/protected
   - Provide getter/setter methods to access/update them

3. Implement **Inheritance**:
   - Create two derived classes:
     - `VegItem`
     - `NonVegItem`

4. Demonstrate **Polymorphism**:
   - Override `calculate_total()` in both classes:
     - VegItem → no extra charge
     - NonVegItem → 10% extra charge

5. Add functionality:
   - Method `place_order(quantity)`
   - Calculate total bill using price × quantity
   - Apply additional charge for non-veg items

6. Implement **Exception Handling**:
   - Use `try-except-finally`
   - Handle:
     - invalid quantity (negative or zero)
     - invalid price
   - Create a **User-Defined Exception**:

     - `InvalidOrderException`
     - Raise it when:

        1. quantity ≤ 0
        2. price ≤ 0
    - Use **Exception Chaining**:
        - Raise `InvalidOrderException` from original exception

7. Main Program should:
    - Create one VegItem and one NonVegItem
    - Place orders
    - Display total bill
    - Handle exceptions
    - Use finally block for completion message

Question: 2 [10 marks]

Weather Report Analysis System

A weather monitoring center wants to analyze weather records. Write a Python program using **NumPy, Pandas, and Matplotlib** to perform weather data analysis.

The dataset contains the following details: **Day, Temperature, Humidity, Rainfall, and Wind Speed**.

Your program should:

1. Create **NumPy arrays** for at least **5 days** and compute the **average**, **highest**, and **lowest** values of the weather parameters.
2. Calculate the **total rainfall** using NumPy.
3. Create a **Pandas DataFrame** with the columns:
   `Day, Temperature, Humidity, Rainfall, Wind_Speed`
4. Add suitable calculated columns such as **Average Weather Reading** and **Rainfall Status**.
5. Filter the days where **Temperature** is above a suitable value and sort the data based on **Temperature** or **Rainfall**.
6. Plot a **bar chart** showing the **Rainfall** of each day.
7. Plot a **line graph** showing the variation in **Temperature** over the days.
8. Display the final DataFrame and graphs with proper **title, labels, and legend**.

----------All the Best----------